JOHN DEERE

John Deere Construction & Forestry: Sand Leveling Challenge

Background

At Deere we run so life can leap forward. Whether our equipment is used to build a road, prepare a homesite, or shape any critical piece of infrastructure the site must be carefully prepped so the structure can rest on solid, predictable ground.

Mission

Using a robot vehicle kit as your base, build a machine that levels sand to a perfectly flat plane inside a sandbox with a mystery terrain structure. You may augment the kit with extra motors, custom mechanisms, 3D-printed or laser-cut parts, cardboard mockups, and onboard/external compute.

What We Provide

- One robot vehicle kit per team (base chassis + controller).
- Access to a makerspace: 3D printers, laser cutter, hand tools, cardboard, fasteners.
- Sandbox for testing

Team and Format

- Team size: 2 to 4 students. One student non CS
- Autonomy: Not required, but partial and full autonomy earn bonus points.
- Computing: Onboard (Arduino, microcontrollers) and/or offboard (laptop, micro-PC) allowed

Success Criteria & Scoring

Projects will be evaluated on the effectiveness of sand leveling, operational speed, and a technical pitch covering design, safety, and circuitry. Bonus points are awarded for autonomy and site safety, while penalties are incurred for manual repairs or tethering during the trial.